

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Award

Tuesday 23 June 2026

Afternoon (Time: 3 hours)

Paper reference **9811/01**

Advanced Extension Award

Mathematics

You must have:
 Mathematical Formulae and Statistical Tables
 An insert for questions 5, 6 and 7 (enclosed)

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P81302A

©2026 Pearson Education Ltd.
P:1/1/1/1/



1. $y = \operatorname{cosec} x \quad x \in \mathbb{R} \quad x \neq n\pi \quad n \in \mathbb{Z}$

where x is measured in radians.

By writing $\operatorname{cosec} x = \frac{1}{\sin x}$, use differentiation from first principles to prove that

$$\frac{dy}{dx} = -\operatorname{cosec} x \cot x$$

You may

- use without proof the formulae for $\sin(A \pm B)$
- assume that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$

You may **not** use without proof

- the chain rule
- the product rule
- the quotient rule

(6)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2.

$$f(x) = x^2 - 3x + 5 \quad x \leq k$$

where k is a constant.

Given that $f(x)$ has an inverse function,

- (a) determine the maximum possible value of k . (2)

For the value of k found in part (a),

- (b) determine $f^{-1}(x)$, stating the domain of f^{-1} (3)

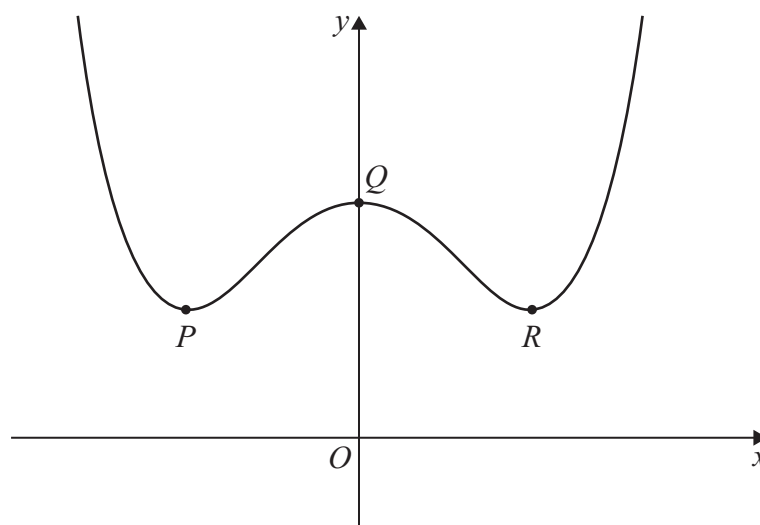


Figure 1

Figure 1 shows a sketch of the curve with equation $y = g(x)$, where

$$g(x) = x^4 - 3x^2 + 5$$

The points P , Q and R , also shown in Figure 1, are the turning points of the curve with equation $y = g(x)$.

Given that

$$h(x) = x^4 - 3x^2 + 5 \quad 0 \leq x \leq a$$

where a is the x coordinate of R ,

- (c) use the answer to part (b), or otherwise, to determine $h^{-1}(x)$, stating the domain of h^{-1} (2)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. A discrete random variable X takes positive integer values only.

The probability distribution of X is given by

- $P(X = 1) = \frac{1}{3}$
- $P(X = 2) = a$

where a is a constant, $0 < a < 1$, and

- if k is an **odd** integer and $k > 1$, then $P(X = k) = \frac{1}{9}P(X = k - 2)$
- if k is an **even** integer and $k > 2$, then $P(X = k) = \frac{1}{\sqrt{5}}P(X = k - 2)$

This is illustrated in the table below for the first six values of x

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{3}$	a	$\frac{1}{27}$	$\frac{1}{\sqrt{5}}a$	$\frac{1}{243}$	$\frac{1}{5}a$

- (a) Determine, in terms of a where appropriate,

(i) $P(X = 7)$

(ii) $P(X = 8)$

(1)

- (b) Show that

$$a = \frac{5 - \sqrt{5}}{8}$$

(2)

- (c) Determine $P(1 < X \leq 8 \mid X \text{ is even})$, giving the answer as a rational number.

(3)

Three independent observations of X are made, denoted by X_1 , X_2 and X_3

The variable Y is defined as $Y = X_1 + X_2 + X_3$

- (d) Determine $P(Y < 5)$, giving the answer in simplest form.

(3)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4.

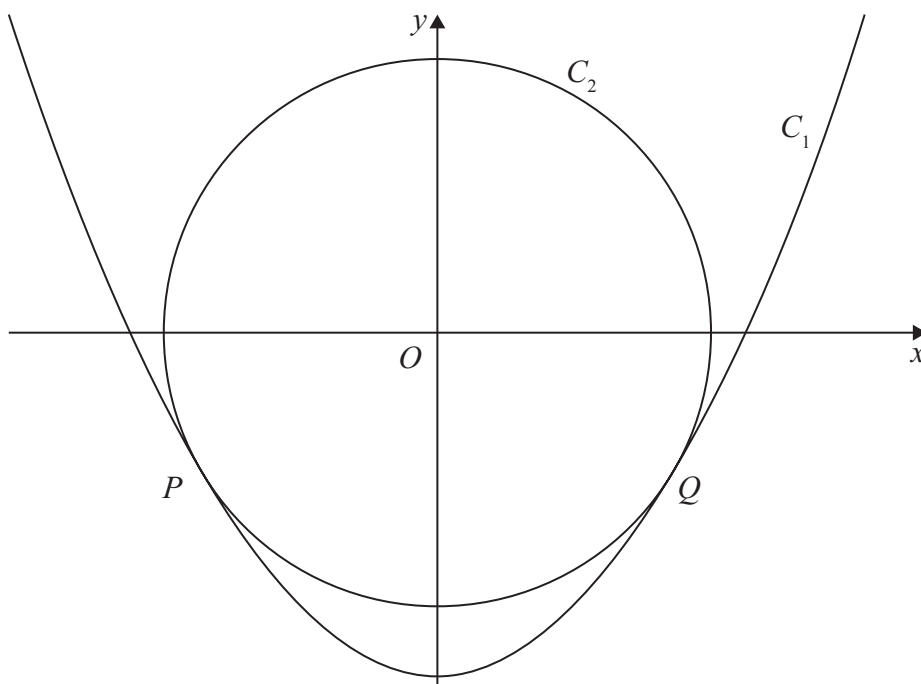


Figure 2

The curve C_1 , shown in Figure 2, has equation

$$y = Ax^2 + k$$

where A and k are constants with $A > 0$ and $k < 0$

The circle C_2 , also shown in Figure 2, has centre at the origin and radius r , where r is a positive constant.

Given that C_1 and C_2 touch at exactly two distinct points, P and Q ,

(a) (i) show that

$$k = -\frac{1}{4A} - Ar^2 \quad (4)$$

(ii) Hence determine the y coordinates of the points P and Q in terms of A only. (2)

(iii) Show that the x coordinates of the points P and Q are given by

$$x = \pm \sqrt{r^2 - \frac{1}{4A^2}} \quad (1)$$

(iv) Determine, in terms of A , the range of possible values of r . (1)



- (b) (i) determine an equation for C_3
Give your answer in the form $y = f(x)$ in terms of A , a , b and r (2)
- (ii) determine the equations of all lines that are normal to both C_3 and C_4
Give the equations in terms of A , a , b and r where appropriate. (4)
- (+S1)

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 4 is 15 marks)



5.

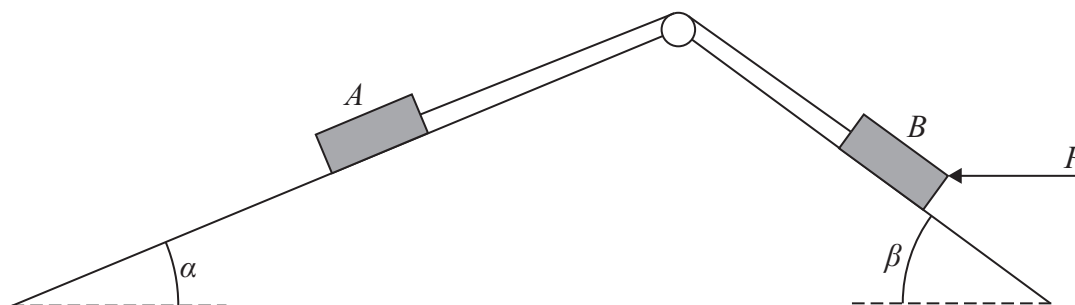


Figure 3

Figure 3 shows a block A with mass $26m$ and a block B with mass $75m$

Block A lies on a rough plane inclined at an angle α to the horizontal.

Block B lies on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth fixed light pulley.

The pulley lies on the line where the two inclined planes meet, with A , B and the pulley all lying in the same vertical plane.

The part of the string between A and the pulley lies along a line of greatest slope of the plane of angle α

The part of the string between B and the pulley lies along a line of greatest slope of the plane of angle β

The two blocks, A and B , are modelled as particles.

- $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{3}{4}$
- the coefficient of friction between block A and its plane is 0.5
- the coefficient of friction between block B and its plane is 0.2

A horizontal force of magnitude P acts on block B . The line of action of P lies in the same vertical plane as A , B and the pulley. The tension in the string is T .

When $k_1 mg \leq P \leq k_2 mg$, the system is at rest in equilibrium with the string taut.

- (a) Determine the exact values of k_1 and k_2 (9)

Assuming that the system remains in equilibrium,

- (b) determine, in simplest form in terms of T , the reaction force acting on the pulley. (3)

Question 5 continued

Without further calculation,

(c) explain the effect of decreasing P below $k_1 mg$ on

- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)

Without further calculation,

(d) explain the effect of increasing P above $k_2 mg$ on

- (i) block B
- (ii) block A
- (iii) the reaction force on the pulley

Justify your answers.

(3)
(+S2)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 20 marks)



6. Given that

$$y = \arctan x$$

(a) prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad (2)$$

Given that f is a function such that the integrals exist,

(b) (i) use the substitution $u = a + b - x$ to show that

$$\int_a^b f(x) \, dx = \int_a^b f(a + b - x) \, dx$$

where a and b are constants

(ii) determine the value of the constant λ such that

$$\int_a^b f(x) \, dx = \lambda \int_a^b (f(x) + f(a + b - x)) \, dx \quad (3)$$

Given that $I = \int_{-\pi}^{\pi} \frac{x(1 + \sin x)}{1 + \cos^2 x} \, dx$

(c) use detailed reasoning to show that

$$I = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx \quad (2)$$

Given that

$$g(x) = \frac{x \sin x}{1 + \cos^2 x}$$

(d) show that

$$g(x) = g(-x) \quad (1)$$

(e) Using the results from parts (a), (b), (c) and (d), determine the value of I

(8)
(+S2)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 18 marks)



7. $y = x^x \quad x > 0$

By taking logarithms of both sides, or otherwise,

- (a) determine $\frac{dy}{dx}$, giving the answer in simplest form in terms of x only. (3)

Given that, for $x > 0$

$$f(x) = (x^x)^x$$

$$g(x) = x^{(x^x)}$$

- (b) determine the exact coordinates of the turning point of $y = f(x)$ (5)
- (c) Show that $g'(x)$ can be written in the form

$$g'(x) = P(x)x^{Q(x)}$$

where $P(x) = x(\ln x)^2 + x \ln x + 1$ and $Q(x)$ is a function of x to be found. (4)

- (d) Solve $P(x) = 1$ (2)

Given that $y = P(x)$ has a stationary point at $x = \alpha$ and a stationary point at $x = \beta$, where $\alpha < \beta$,

- (e) determine the value of α and the value of β (3)
- (f) Hence show that $P(x)$ has a minimum at $x = \beta$ (2)

Using the answers to parts (c), (d), (e) and (f),

- (g) show that $g(x)$ is an increasing function for $x > 0$ (4)
(+S2)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



P 8 1 3 0 2 A 0 3 9 4 4

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 25 marks)

TOTAL FOR PAPER IS 100 MARKS

